MPX · PD-298 · DESIGN REVIEW

MPx Spin v1

Spin the water through a filter sleeve, then read the sleeve

Concept 04 developed from our centrifuge experiment. What it does, what the calibrated simulation shows, which of our 22 requirements it meets, and the **contradictions, trade-offs and failure modes** to take into TRIZ and FMEA.

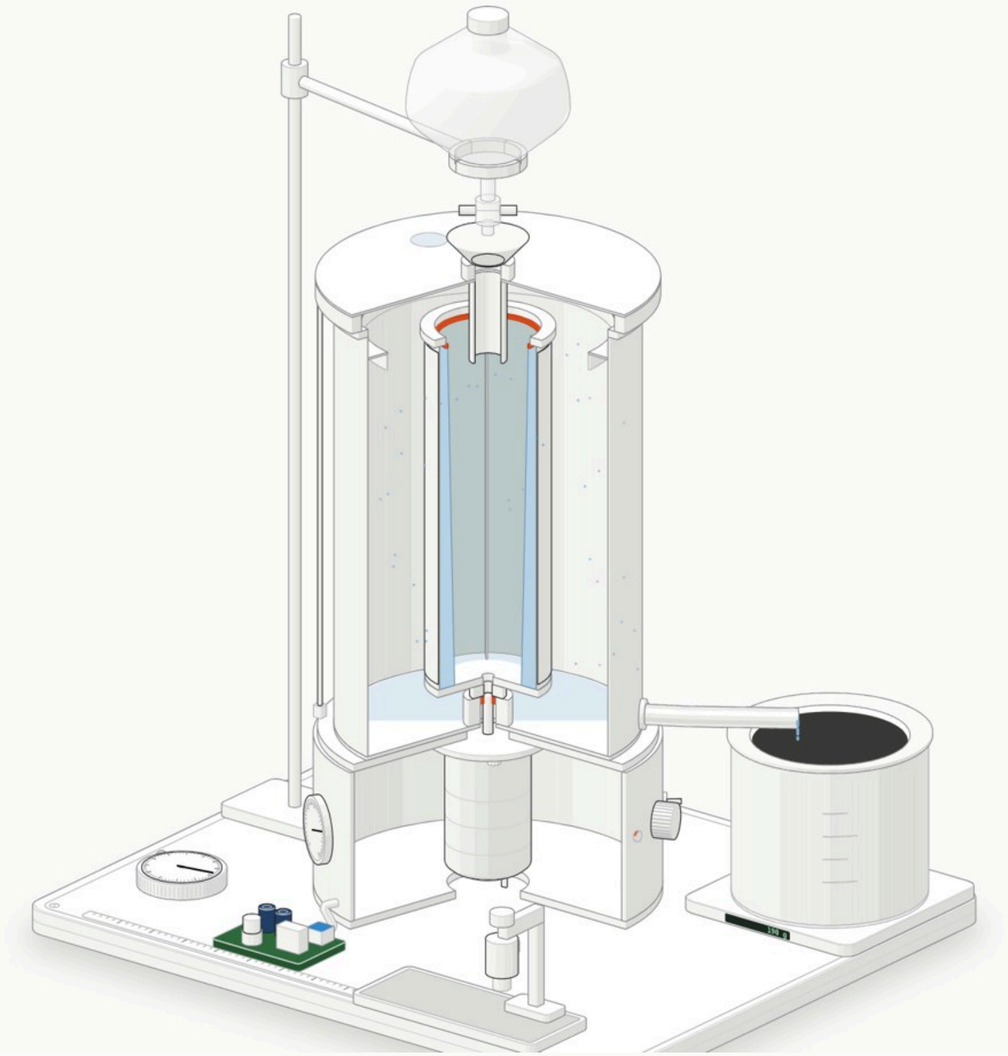
28 L/h
per 400 mL batch cycle, target 15*

94 %
caught by count, 1–100 µm*

71 L
per sleeve on tap water*

6.5 %
how fast the rig's water really turned

* simulated: model calibrated on the team's 3-speed rig; vanes assumed to turn the water at 85 % of basket speed



The simulated test bench: basket at 1,500 rpm, water held against the sleeve

CONCEPT DEVELOPMENT · OCT 2026

Where this concept sits in the CED process

Concept 04 was one of the three concepts we carried out of evaluation. The rig gave us real data, so this deck turns it into an embodiment we can **attack with TRIZ and FMEA**.

LEC 3–5 ✓
Task clarification
Requirements list (22), SNPS, objective tree

LEC 6–8 ✓
Conceptual design
Concept 04: paper + mesh + centrifuge + carbon

RIG ✓
Physical test
200 mL, motor off / 3,500 / ≈5,000 rpm, 3 trials each

LEC 9 ✓
Simulation
Model calibrated on the rig (this deck)

NOW
Embodiment
Spin v1 layout, teardown, requirement check

NEXT
TRIZ + FMEA
Contradictions and failure modes set up on slides 17–18

LATER
Detail design
After bench tests 1–8

EMBODIMENT DESIGN STEPS (LECTURE 2, SLIDE 36) AND WHERE SPIN V1 STANDS

✓ Overall layout: arrangement, spatial compatibility	Single dimension set (dims.mjs) drives the bench simulation and the teardown
✓ Preliminary form design: shapes, materials, processes	316L basket and mesh, 304 bowl and lid, cellulose sleeve, PC/ABS dry base
◑ Apply all major criteria	Flow, capture, sleeve life, stress by simulation; cost, noise, drop open
◑ Develop with scale drawings and critical reviews	Exploded + cutaway drawings; this deck is the critical review
▶ **Identify errors and disturbing factors**	TRIZ contradictions and 18 FMEA candidates, unrated (slides 17–18)
○ Optimise	Speed, paper grade, vane design, after bench tests
○ Outcome: embodiment ready for detail design	Not yet

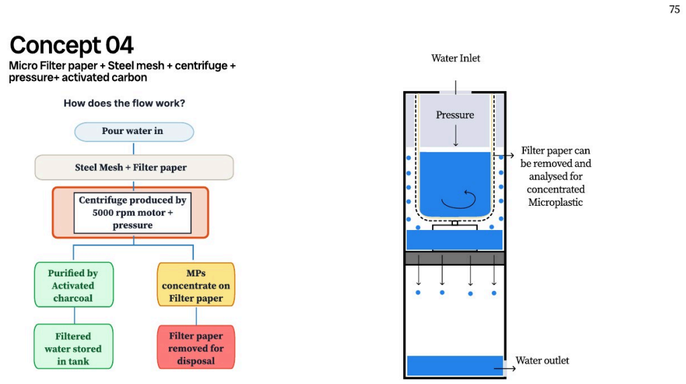
THREE THINGS THE RIG TAUGHT US

- **The water hardly spun.** Fitting the model to the three speeds says the water turned at **6.5 %** of the motor: about 3 g, not 700 g.
- **The paper was too coarse to count.** A 25 µm paper catches about **12 %** of 1–100 µm particles by number.
- **The 106 L/h is a drain rate,** not a flow. With pouring and stopping, the rig's cycle is about 33 L/h.
- So Spin v1 adds **vanes**, a **< 2 µm cellulose sleeve** and a slower, controlled **1,500 rpm**.

Sources: CED lec02 sl. 36–37; MPx_01 (Concept 04 + experiment, pp. 75–79); model.mjs

What it does: from Concept 04 to Spin v1

Pour a batch in, spin it, and the water is driven out through a filter sleeve. The microplastics stay on the sleeve, which then becomes the sample under the microscope.



Concept 04 as selected (MPx_01, p. 75)

WHAT CHANGED, AND WHY

≈5,000 rpm 775 motor	→	**1,500 rpm 24 V BLDC + 4 vanes** water must turn with the basket
25 µm paper	→	**< 2 µm cellulose sleeve** 25 µm catches 12 % by count
plastic frame, base	→	**316L / 304 wet path** no plastic shed into the water
activated carbon tank	→	**optional carbon cup at the spout** keep the filter step clean to sample
open top	→	**lid + interlock switch** no spin with the lid off

1 Pour

Fill **400 mL** through the lid chute while the basket starts. The four vanes catch the water and bring it up to **85 %** of basket speed (assumed).

2 Spin

At **1,500 rpm** the water stands as a wall against the sleeve at about **55 g**. The wall pressure reaches **7 kPa**, four times the head of a full basket at rest.

3 Filter

Water passes the **cellulose sleeve**, the **50 µm** 316L mesh and the **perforated cage**, is flung onto the bowl and leaves by the spout. 90 % is through in **28 s**.

4 Read the sleeve

Lift the lid, the clamp ring and the vane insert; pull the sleeve. Stain with Nile red and view at **100–250×** (MP-250) with a phone.

KEY NUMBERS (SIMULATED)

Size	Ø170 × 346 mm; basket Ø60 × 180
Batch	400 mL; 8 batches for 3 L
Flow	28 L/h per cycle (pour, spin, stop)
Capture	94 % by count; 89 % at 1–2 µm
Sleeve	71 L tap, 18 L borewell before < 15 L/h
Drive	24 V BLDC, 1,000–1,750 rpm band
Wet path	316L, 304, cellulose, silicone, FKM
Cost	Not costed yet

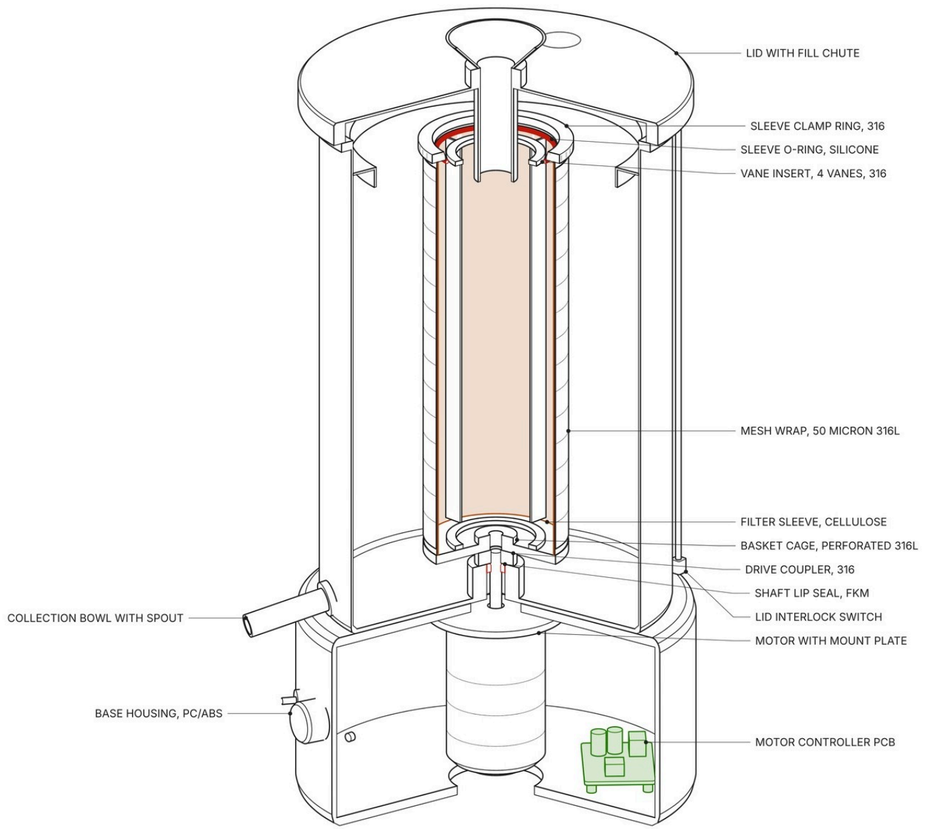
AGAINST OUR SNPS

Remove	Detect	Contain
yes, < 2 µm sleeve	the sleeve is the sample	sleeve bagged whole

Concept 04 from MPx_01 p. 75; Spin v1 dimensions dims.mjs; numbers from model.mjs at 1,500 rpm, 25 °C, tap water

How it works: cut open

Everything wet is steel or cellulose; the motor and the board sit in a dry base under the bowl. A lip seal and the bowl floor keep them apart.



WHERE THE WATER GOES

1 → 2 → 3

Lid chute
batch in, Ø16 bore, funnel

Vane insert
4 radial vanes drag the water round

Cellulose sleeve
the filter and the sample, < 2 µm class

4 → 5 → 6

50 µm 316L mesh
supports the wet paper over the holes

Perforated cage
Ø2 holes at 40 % open, 0.6 mm 316L

Bowl and spout
304, splash skirt, out to a jug

WHY IT FILTERS FASTER WHEN IT SPINS

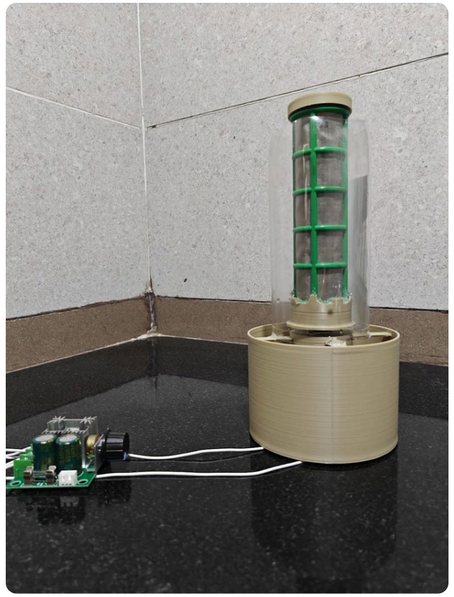
$$z(r) = z_0 + \omega^2 r^2 / 2g \qquad p_{wall}(z) = \rho\, g\, (z_R - z)$$

The spinning water takes a paraboloid shape. Where it meets the sleeve it stands much higher than the basket is tall, so the pressure across the paper rises from **1.8 kPa** (still, full) to **7.2 kPa** at 1,500 rpm with vanes. Flow through the wet wall follows Darcy's law with the paper's permeance. Only the water's own speed counts: if it lags the basket, the head collapses.

Teardown drawing from dims.mjs (cutaway at 110°); illustrative concept, not for manufacture

The experiment: centrifuge-assisted filtration

Does spinning a cylindrical filter make water pass faster than gravity? 200 mL per trial, three trials at three conditions (MPx_01, pp. 76–79).



The rig: Ø50 × 150 mm cage, PWM board

RESULTS, TIME FOR 200 ML TO PASS (S)

Condition	Trial 1	Trial 2	Trial 3	Mean	SD	Apparent L/h
A · motor off	20.02	24.77	22.49	**22.43**	2.38	32.1
B · 3,500 rpm	10.14	10.75	10.16	**10.35**	0.35	69.6
C · ≈5,000 rpm	7.19	6.55	6.58	**6.77**	0.36	106.3

Spinning cut the time from 22.4 s to 6.8 s: **3.3×** the baseline. The team's notes: RPM was not verified, removal and water quality were not measured, loading and clogging not tested.

READING THE NUMBERS AGAIN

- **106 L/h is how fast one batch drains,** not what a user gets. Add the pour and the stop and the rig's cycle is about **33 L/h**.
- **3.3× faster at 5,000 rpm is far too little.** If the water turned with the basket, the pressure at the wall would be about 60× the static head.
- **A 25 µm paper is a pre-filter** for microplastics. By count most particles are under 20 µm; it catches about 12 %.
- **SD at motor-off is 2.4 s** (11 %): the start and stop of the timing matters as much as the speed.

METHOD

1. Add 200 mL to the filter cylinder
2. Time until the water has passed and collected below
3. Motor off, 3,500 rpm, ≈5,000 rpm (PWM)
4. Three readings per condition
5. Mean time and flow rate

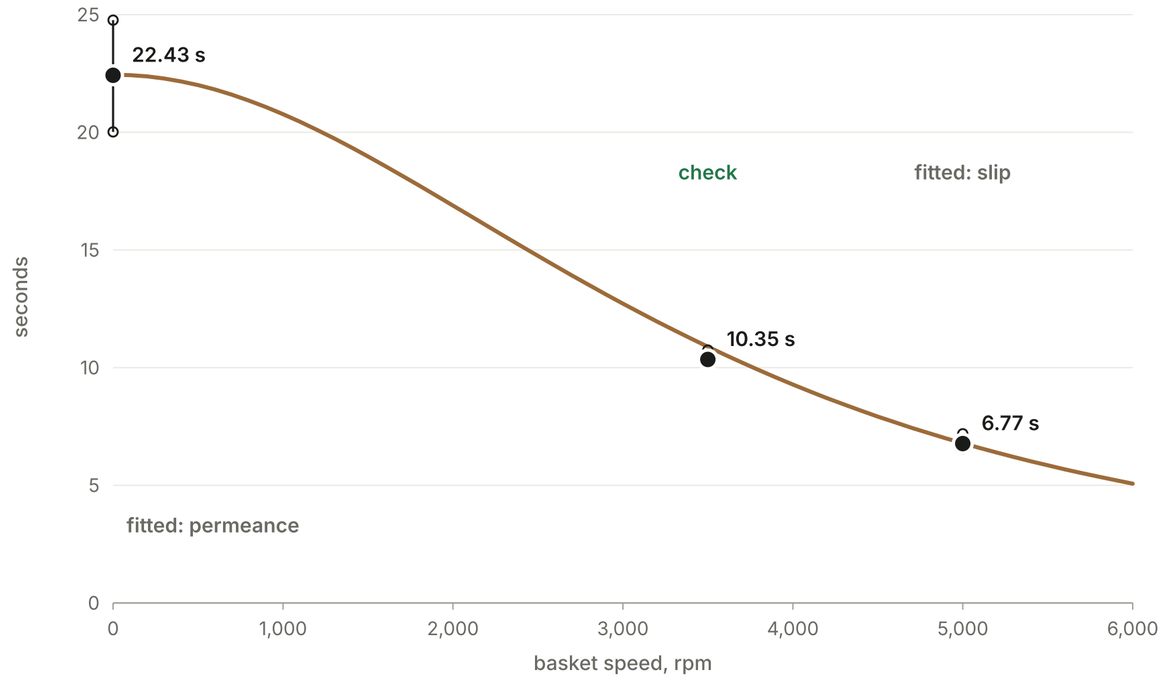
Kit: 12 V 775 motor (3,500–9,000 rpm rated), 50 µm steel mesh cylinder in a plastic frame, 25 µm paper lining, printed housing, PWM controller.

Data and photo: MPx_01, pp. 76–79. Apparent L/h = 200 mL ÷ mean time

Calibrating the model on the rig: the water hardly turned

The model has two unknowns. The motor-off run fixes the paper's lumped permeance, the ≈5,000 rpm run fixes how fast the water turns. The 3,500 rpm run is then a check.

Time for 90 % to pass vs basket speed, rig (200 mL, Ø50 × 150, 25 µm paper)



Dots: the three trials and their mean. Line: the model with $\kappa = 8.93 \times 10^{-9}$ m and the water at 6.52 % of motor speed. Model 22.45 / 10.89 / 6.79 s: it predicts **10.9 s** at 3,500 rpm against **10.35 s** measured (+5 %).

WATER SPEED

6.5 %

of motor speed. At 5,000 rpm the water turned at about **326 rpm**: **3 g** at the wall instead of about 700 g. A smooth basket and a 7 s run never let it catch up.

LUMPED PERMEANCE

3.2×

the Whatman Grade 4 datasheet value. Either the paper was not the only path (seam, ends, frame gaps) or it was not seated. **Bypass would let particles through.**

WHAT THIS CHANGES

vanes

The design has to make the water turn. Spin v1 assumes four vanes reach 85 %; the model needs at least **55 %** to make 15 L/h at 1,500 rpm.

Fit: fit.py / fit2.py (least squares on means). 90 % pass time used as the measured end point

The calculations behind the numbers

Eight hand-checkable relations from model.mjs. Spin v1 values at 1,500 rpm, 400 mL, 25 °C, Grade 602 h class sleeve, tap water.

1 Wall pressure

$$p_{wall} = \rho\, g\, z_R\,,\quad z_R = z_0 + \omega^2 R^2/2g$$

solid-body rotation, water at 0.85 × 157 rad/s, R = 30 mm; 400 mL wall

7.2 kPa

vs 1.8 kPa at rest: 4× the head.

2 Flow through the wall

$$Q = K \cdot 2\pi R \cdot \rho g\,(z_R h - h^2/2)$$

Darcy through the wetted height h, base closed

118 L/h

at the start of the batch; falls as it drains.

3 Paper permeance

$$K = V / (t_H\, A\, \Delta p) \cdot \mu_{20}/\mu_T$$

Herzberg: 100 mL, 10 cm², 10 cm head; 602 h: 750 s

1.53×10^{-7} m/(s·Pa)

rig lumped: 1.0×10^{-5} (Grade 4 datasheet ×3.2).

4 Batch cycle

$$Q_{cycle} = 0.9V / (t_{pour} + t_{90} + t_{stop} + 5\ s)$$

10 s + 28.2 s + 3 s + 5 s for 360 mL

28 L/h

the number to hold against 15 L/h.

5 Capture by count

$$\eta(d) = 1 / (1 + (d_{50}/d)^4)\,,\quad n \propto d^{-2}$$

98 % at the paper's rating; 1–100 µm particles

94 %

89 % at 1–2 µm; Grade 4: 12 %.

6 Face velocity

$$v_{face} / v_{test} = (Q / A_{wet}) / (100\ mL / 10\ cm^2\ t_H)$$

how much faster than the rating test the paper works

7.2×

we allow ≤ 10× (assumption).

7 Cake on the sleeve

$$1/K = 1/K_0 + \mu\, \alpha\, m / A\,,\quad \alpha = 10^{13}\ m/kg$$

solids 0.5 mg/L tap, 2 borewell, 10 turbid

71 L

tap, until the cycle drops below 15 L/h.

8 Basket stress

$$\sigma = \rho_s(\omega R)^2 + pR / t(1-\varphi)$$

316L, 0.6 mm wall, 40 % open; yield 170 MPa

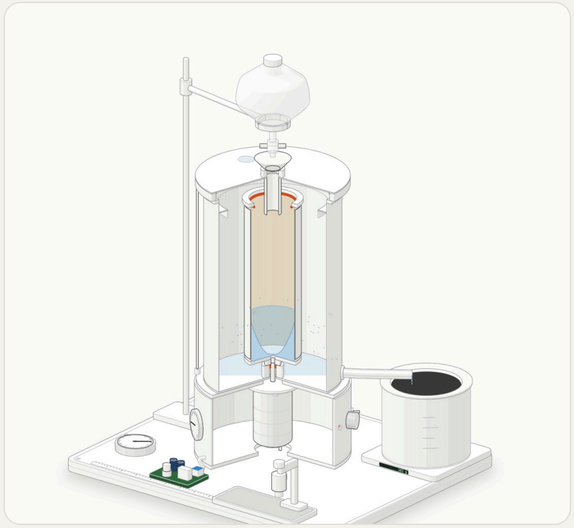
0.78 MPa

safety factor 219×; 5 g off-centre: 3.7 N.

Inputs: dims.mjs, Whatman qualitative datasheet (ratings, Herzberg times), rig calibration; α and the 85 % vane slip are assumptions

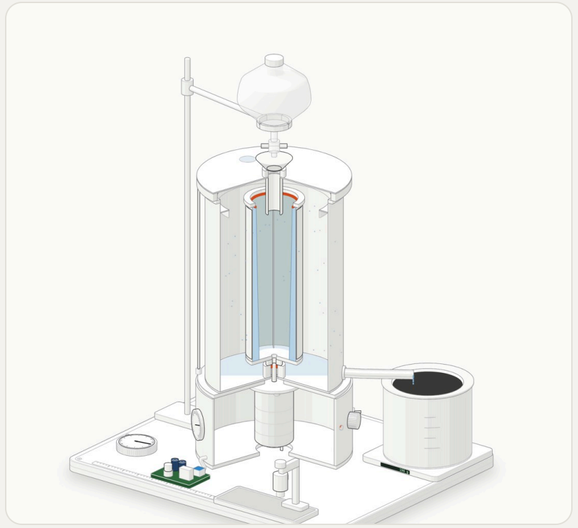
The simulated test bench: four states

An interactive bench driven by the same model: separatory funnel doses the batch, the spout pours through a carbon cup into a jug on a scale, a stopwatch times it, the sleeve goes under the microscope.



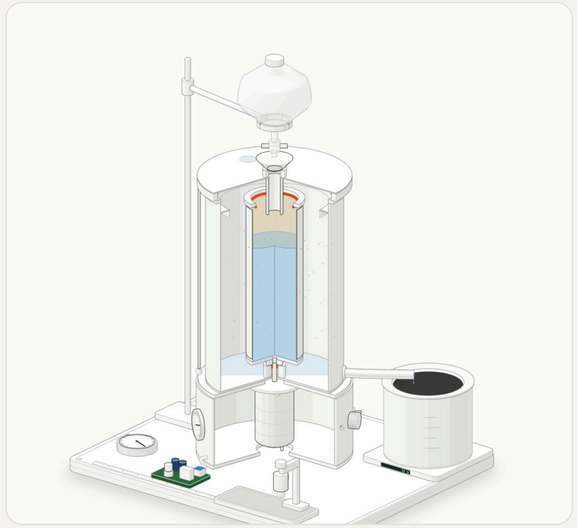
A · The experiment, ≈5,000 rpm

The rig as calibrated. The motor is at 5,000 rpm but the water turns at 326 rpm: it slumps in the basket and drains slowly.



B · Spin v1, 1,500 rpm

Vanes on. The water stands against the sleeve at 55 g and is through in about 28 s per 400 mL.



B · Spin v1 without vanes

Same basket and sleeve, vanes removed: the water lags like the rig's and the fine paper passes only 1.2 L/h.



C · Lift the sleeve

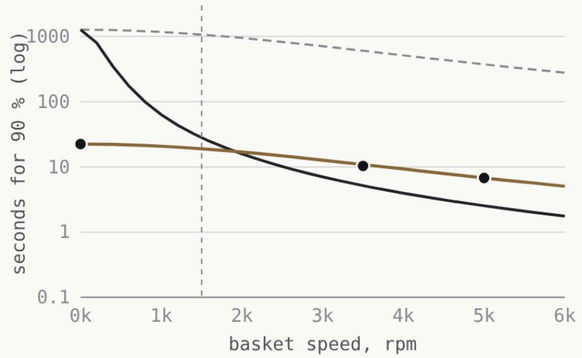
Lid aside, basket out, sleeve flattened on the tray under the pocket microscope.

Open it: Desktop › MPx › Centrifuge_v1 › figure › MPx_Spin_v1_TestBench.html. Sliders for speed, batch, temperature; paper grade, vanes, water and the experiment's three conditions.

What the bench reads out

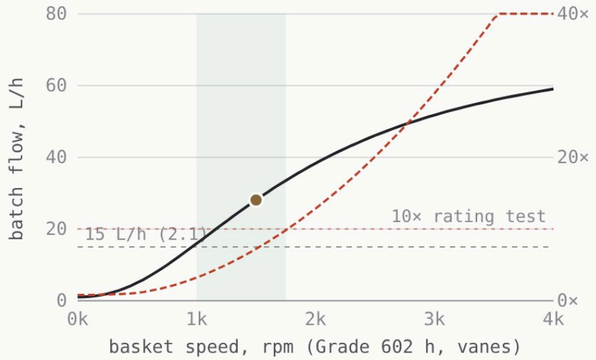
Two live charts, the requirement checks for the current settings, and the microscope's view of the sleeve.

Time for 90 % to pass, by speed



● measured (3 trials) — rig model, 200 mL — v1, Grade 602 h, vanes -- v1 without vanes, 400 mL.

Choosing the speed



— flow per batch cycle -- face velocity ÷ the paper's rating test (right axis). Green: both hold from **1,000** to **1,750 rpm**.

Pocket microscope on the sleeve

The sleeve is 188 × 180 mm of paper: a big slide. **207** particles of 1–100 µm are on it after 0.2 L, about **0.003** per 0.8 mm field, so a count of 100 would need about **32,649** fields. Concentrate before counting: run 20 L or more onto one sleeve, or cut a small disc from it.

Cellulose takes up Nile red too, so the paper glows faintly: a blank sleeve sets the background to subtract.



250× · field 0.8 mm · Nile red, blue LED. Representative mix; **9** resolvable, 32 under 5 µm.

SPIN V1, 1,500 RPM, VANES, GRADE 602 H

Requirements, live

Microplastics caught by count, 1–100 µm	94 %	≥ 90 %	6.1	pass
Caught at 1–2 µm	89 %	≥ 90 %	6.1	fail
Flow per batch cycle, 400 mL	28 L/h	≥ 15 L/h	2.1	pass
Face velocity vs the paper's rating test	7×	≤ 10×	6.1	pass
Litres per sleeve before flow < 15 L/h (tap)	71 L	≥ 20 L	9.1	pass
Water speed vs basket speed	85 %	≥ 70 %	2.1	pass
Basket stress safety factor	219×	≥ 4×	3.5	pass
Plastic in the wet path	none	none	3.2	pass
Paper is the sample for the microscope	yes	yes	1.2	pass
Lid interlock before spin	switch	fitted	safety	pass

THE EXPERIMENT'S RIG AT ≈5,000 RPM

Requirements, live

Microplastics caught by count, 1–100 µm	12 %	≥ 90 %	6.1	fail
Caught at 1–2 µm	0 %	≥ 90 %	6.1	fail
Flow per batch cycle, 200 mL	33 L/h	≥ 15 L/h	2.1	pass
Face velocity vs the paper's rating test	2×	≤ 10×	6.1	pass
Litres per sleeve before flow < 15 L/h (tap)	1.8 L	≥ 20 L	9.1	fail
Water speed vs basket speed	7 %	≥ 70 %	2.1	fail
Basket stress safety factor	117×	≥ 4×	3.5	pass
Plastic in the wet path	frame + base	none	3.2	fail
Paper is the sample for the microscope	yes	yes	1.2	pass
Lid interlock before spin	no lid	fitted	safety	fail

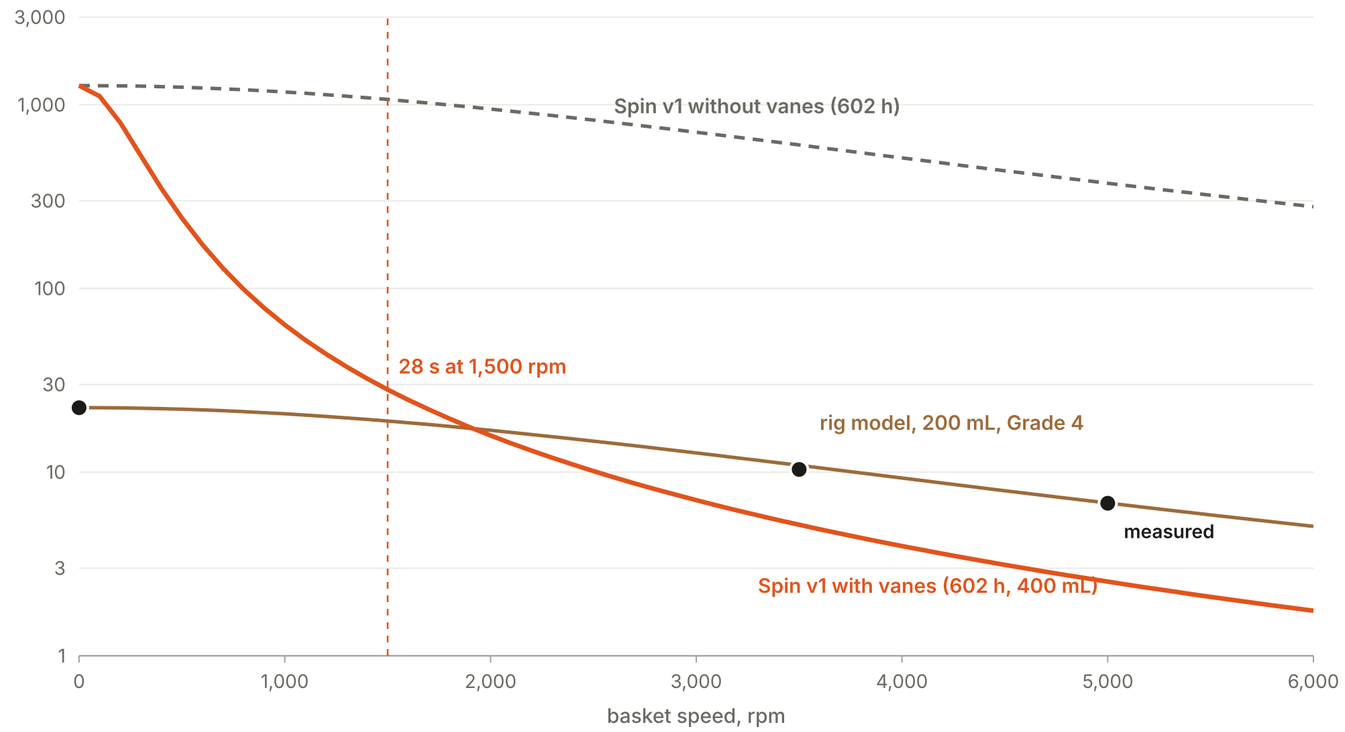
The rig passes flow, face velocity and stress, but fails capture, sleeve life, water speed, the wet-path rule and the interlock. Spin v1 passes all but the 1–2 µm band (89 %).

Screenshots from MPx_Spin_v1_TestBench.html

Result 1 · Vanes decide whether the concept works at all

With a fine sleeve, the head from a lagging water ring is far too low. Without vanes Spin v1 passes 1.2 L/h; with them, 28 L/h.

Time for 90 % of the batch to pass vs basket speed (log)



Spin v1 uses a paper 20× tighter than the rig's (Herzberg 750 s vs 37 s). Without vanes that paper would need minutes per batch at any speed.

HOW MUCH WATER SPEED IS ENOUGH? (1,500 RPM)

Water / basket	Cycle L/h	Face ×	Sleeve L
20 %	2.9	0.9	—
30 %	5.5	1.2	—
50 %	13.0	2.5	—
70 %	21.8	4.9	32
85 %	28.1	7.2	71
100 %	33.8	10.0	118

Below about **55 %** the cycle falls under 15 L/h. 6.5 % (the rig) gives about 1 L/h.

WHY THE WATER LAGS, AND WHAT VANES DO

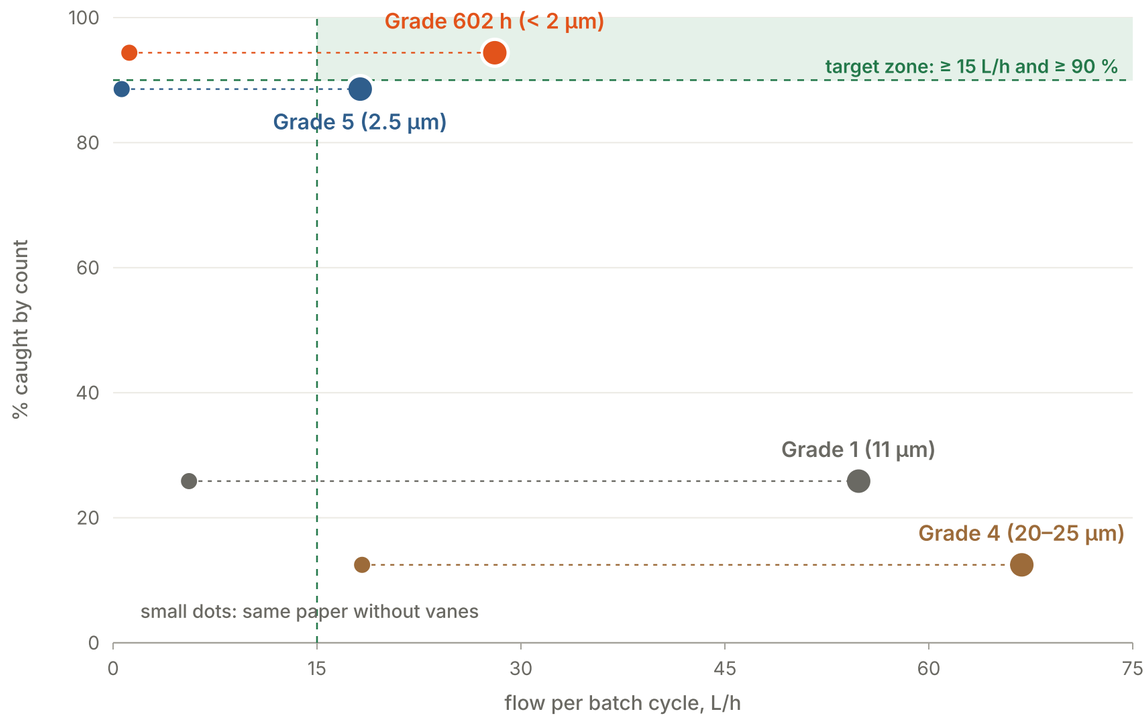
- A smooth wall only drags the water by viscosity; spin-up takes far longer than a 7–30 s batch.
- Radial vanes push the water round like paddles, so it reaches basket speed in about a revolution.
- The 85 % figure is an assumption: **bench test 1** measures it.

model.mjs; vane slip varied by scaling the water speed at a fixed 1,500 rpm basket

Result 2 · Paper grade: catching small particles costs flow

Four plain cellulose papers in the same Spin v1 at 1,500 rpm with vanes. Finer papers catch far more by count, but pass less water and clog sooner.

Capture by count (1–100 µm) vs batch-cycle flow



Grade 4 is the rig's paper. The < 2 µm Grade 602 h class is the only one inside the target zone; Grade 5 (2.5 µm) is tighter in Herzberg time but rated coarser, so it catches less and flows less.

THE FOUR PAPERS AT 1,500 RPM WITH VANES

Paper	1–2 µm	all	L/h	Face ×	Sleeve L
Grade 4	0 %	12 %	67	7.2	119
Grade 1	2 %	26 %	55	7.2	111
Grade 602 h	89 %	94 %	28	7.2	71
Grade 5	78 %	89 %	18	7.2	27

Face × is the same for all four: it is just the wall pressure divided by the rating test's 10 cm head, whatever the paper. Sleeve L is tap water until the cycle falls below 15 L/h.

THE CONTRADICTION TO CARRY INTO TRIZ

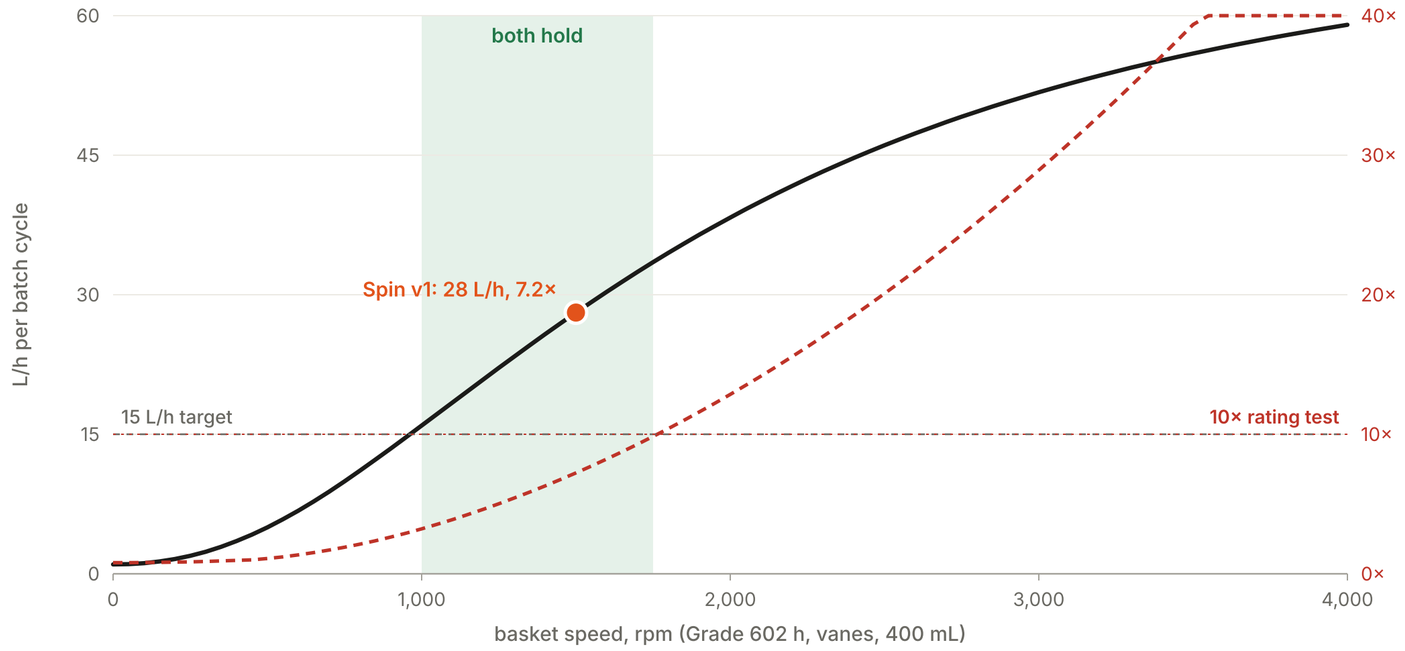
Catching **1–2 µm** particles needs a tighter paper, which cuts flow and sleeve life. Spin v1 sits at **89 %** in that band: just short of 90 %. Going tighter means going faster, a bigger sleeve or a two-stage sleeve.

Whatman qualitative papers: rating at 98 % retention, Herzberg time. Capture curve and count distribution ($n \propto d^{-2}$) are model assumptions

Result 3 · The speed band: 1,000–1,750 rpm

Faster means more flow and a longer-lasting sleeve, but the paper then works far above the velocity it was rated at, so its capture is no longer known.

Batch-cycle flow (left) and face velocity ÷ rating test (right) vs speed



Below 1,000 rpm the cycle drops under 15 L/h; above 1,750 rpm the paper works more than 10× faster than in its rating test. Black: flow. Red dashed: face velocity, right axis.

SPEED VS FLOW, SLEEVE LIFE AND STRESS

rpm	L/h	Face ×	Sleeve L	In band
800	11	2.1	—	no
1,000	16	3.2	4	yes
1,200	21	4.6	28	yes
1,500	28	7.2	71	yes
1,700	32	9.3	106	yes
2,000	38	12.9	165	no
2,500	46	20.1	286	no

AT 1,500 RPM THE SPINNING PARTS ARE EASY

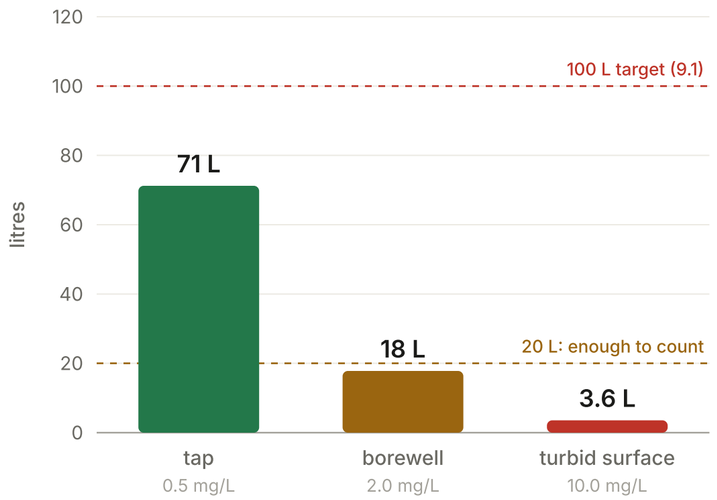
- Basket hoop stress **0.78 MPa**: safety factor **219×** on 316L.
- Water at **55 g**, basket rim at 75 g.
- Unbalance: 1 g off-centre → **0.74 N**, 5 g → **3.7 N**. An uneven wet sleeve is the real vibration source.

model.mjs; the 10× face-velocity limit is our assumption until bench test 5 measures capture at speed

Result 4 · Dirty water, light plastics and a sparse sample

Three effects that the flow numbers alone hide. Each one is a trade-off or a failure mode to carry forward.

LITRES PER SLEEVE BEFORE THE CYCLE < 15 L/H



The cake of silt on the sleeve, not the microplastics, ends its life. Tap water lasts 71 L; borewell water 18 L; turbid surface water about 4 L. **Silt resistance α = 10^{13} m/kg is assumed** (bench test 4).

RADIAL DRIFT IN THE SPINNING WATER, 1,500 RPM

Polymer	ρ kg/m³	10 µm	50 µm
PP	905	-0.26	-6.6
PE	940	-0.17	-4.2
PS	1,050	+0.14	+3.5
PA	1,140	+0.39	+9.7
PET	1,380	+1.06	+26.4
PVC	1,390	+1.08	+27.1

mm/s, + outward. The water itself moves outward through the sleeve at about **0.96 mm/s**. PP above about **19 µm** and PE above about **24 µm** drift inward faster than that: they ride the inner surface and only reach the sleeve as the last water drains, near its top or on the floor. Dense PET and PVC are thrown onto the paper.

READING THE SLEEVE



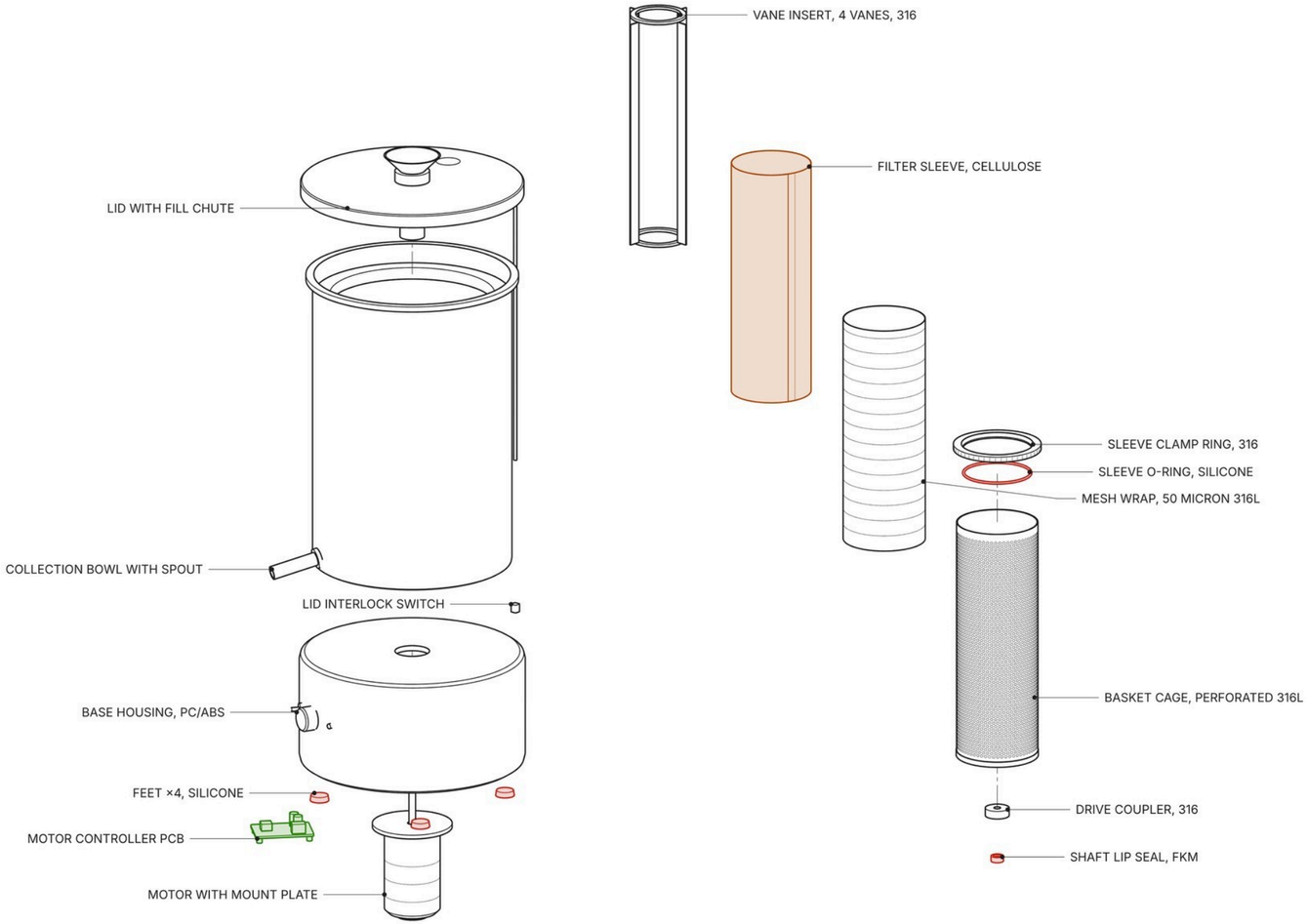
MP-250 at 250×, 0.8 mm field, Nile red (simulated)

- The sleeve is 188 × 180 mm: about 67,000 microscope fields. One 400 mL batch at an assumed 1,000 particles per litre leaves about **0.005 per field**.
- About **20 L** on one sleeve gives ≈ 0.3 per field; or cut and count a small disc, or use a smaller sample sleeve.
- Cellulose takes up Nile red: a **blank sleeve** is needed as background, and cellulose fibres must not be counted as plastic.

Drift: Stokes in a centrifugal field, $v = \Delta\rho\, d^2\, \omega^2\, r / 18\mu$ at r = 25 mm. Microscope: Carson MP-250, 250×, 0.8 mm field

How it goes together

Fifteen parts. The motor and board leave the base from below; the rotor parts lift out of the bowl. The filter stack is mesh outside the cage, sleeve inside, vanes inside the sleeve.



PARTS AND MATERIALS

Lid with fill chute
304, glass window, interlock tab

Collection bowl with spout
304 deep-drawn, splash skirt, boss

Sleeve clamp ring + O-ring
316 + silicone, seals the sleeve top

Vane insert
4 vanes on two hub rings, 316

Filter sleeve
cellulose, < 2 µm class, 190 × 180 mm

Mesh wrap
50 µm woven 316L, on the cage

Basket cage
0.6 mm 316L, Ø2 holes at 40 % open

Drive coupler + lip seal
316 + FKM

Lid interlock switch
on the base deck

Base housing + feet
PC/ABS, dry side; knob, LED, switch

Motor with mount plate
24 V BLDC

Controller PCB
BLDC driver, speed set point

Teardown drawing from dims.mjs; illustrative concept, not for manufacture. Full drawing: figure › MPx_Spin_v1_Teardown.html

Our 22 technical requirements: what Spin v1 meets and what it doesn't

Ranked by our weighted objective tree. **Pass · sim**: shown by the calibrated simulation. **By design**: should meet it, not yet tested.

Pass · sim 3 | Met by design 2 | By design · untested 5 | Partial 5 | At risk 3 | Not yet tested 4

Rank	#	Requirement	Spin v1 result	Checked by	Status
1	**6.1**	Removal ≥ 90 % by count, 1 µm–5 mm (stretch 99 %)	94 % (1–100 µm); 89 % at 1–2 µm; capture at 7× face velocity unverified	Sim	Partial
2	**3.2**	Filter medium sheds no secondary micro/nanoplastic	Steel wet path, no plastic; the cellulose sleeve may shed fibres	Design	By design · untested
3	**6.2**	Detection false +/− ≤ 10 % vs reference microscopy	Sleeve is the sample; Nile red stains cellulose; very sparse fields	—	At risk
4	**10.2**	Captured microplastics stay contained	Caught on the sleeve, which lifts out whole and is bagged	Design	By design · untested
5	**4.1**	Result through ≥ 1 user feedback channel	Microscope view of the sleeve with a phone; nothing on the device	Design	Partial
6	**11.1**	Retail price ₹1,500–6,000	Not costed; 316L basket, 304 bowl and a BLDC drive lead the cost	—	At risk
7	**3.1**	Water-contact parts food-contact safe	316L, 304, cellulose, silicone, FKM	Spec	By design · untested
7	**3.4**	No leachables above IS 10500	Stainless and cellulose; not tested	Spec	By design · untested
9	**2.1**	Flow rate ≥ 15 L/h	28 L/h per cycle at 1,500 rpm with vanes; 1.2 L/h without	Sim	Pass · sim
9	**1.1**	Per-use capacity for 1–3 L/day	3 L = 8 batches of 400 mL, about 6 min	Sim	Pass · sim
9	**10.1**	Disposal pathway for spent filters (W)	Sleeve and catch bagged; no defined route yet	—	Not yet tested
9	**1.2**	Non-destructive sampling (W)	The sleeve itself is the sample	Design	Met by design
9	**5.3**	Display legible outdoors (W)	Status LED only	Design	Partial
14	**9.1**	Filter rated ≥ 100 L or 3 months	71 L tap, 18 L borewell, 4 L turbid	Sim	Partial
14	**7.1**	Filter swap tool-free in ≤ 30 s	Lid, clamp ring, vane insert, sleeve: four steps; not timed	Design	At risk
14	**4.2**	Alert near filter end-of-life	Possible from the batch drain time; not designed	—	Not yet tested
14	**5.2**	Spout parts detachable and washable	Fixed spout on a lift-out bowl	Design	Partial
14	**8.2**	Usable at 5–45 °C	20–35 L/h from 5 to 45 °C (viscosity)	Sim	Pass · sim
19	**3.3**	Housing UV and impact resistant (W)	Steel bowl and lid; PC/ABS base	Design	By design · untested
19	**3.5**	Survives 1 m drop (W)	Basket stress SF 219× at speed; drop not tested	—	Not yet tested
21	**5.1**	Actuation force ≤ 50 N (W)	Knob and button	Design	Met by design
21	**8.1**	Noise ≤ 60 dB(A) (W)	1,500 rpm BLDC plus sleeve unbalance; not measured	—	Not yet tested

Flow (2.1), capacity (1.1) and temperature (8.2) pass in simulation, and removal (6.1) is close: the Tier 1 risks are now detection (6.2), price (11.1) and sleeve life on dirty water (9.1).

Requirements: MPx_Weighted_Objective_Tree.docx (22 items, D/W). W = wish. No requirement fails outright; four are at risk

What the simulation validates, and what it can't

The model reproduces our rig within 5 % and gives the flow and stress side with some confidence. Capture, slip with vanes, cake and detection are assumptions until measured.

CHECKS, SPIN V1 AT 1,500 RPM

Check	Result	Status
Rig prediction at 3,500 rpm	10.9 s vs 10.35 s measured	pass
Flow per batch cycle	28 L/h ≥ 15	pass
Capture by count	94 % ≥ 90	pass
Capture at 1–2 µm	89 % < 90	fail
Face velocity	7.2× ≤ 10×	pass
Sleeve life, tap water	71 L ≥ 20	pass
Sleeve life, borewell / turbid	18 L / 4 L	fail
Basket stress	SF 219×	pass
Usable at 5 °C	20 L/h	pass
Without vanes	1.2 L/h	fail
Sleeve 100 L (9.1)	71 L tap	fail

ASSUMPTIONS TO REPLACE WITH MEASUREMENTS

Assumption	Value used	Drives	Bench test
Water speed with 4 vanes	85 % of basket	Flow (2.1)	1 · strobe or tracer on the water, rig with retrofit vanes
Paper is the only path	lumped K fitted	Capture (6.1)	2 · dye through the seam and ends; K vs datasheet
Capture holds at 7× face velocity	rating curve	Capture (6.1)	5 · spiked 1, 2, 5, 10 µm beads at 0 and 1,500 rpm
Cake resistance of silt	$\alpha = 10^{13}$ m/kg	Life (9.1)	4 · flow vs litres on borewell water
Particle sizes by count	$n \propto d^{-2}$, 1–100 µm	Capture (6.1)	count real tap samples by size
Nile red on cellulose	blank subtracts	Detection (6.2)	6 · stained blank sleeve under the MP-250
Sleeve stays sealed and whole	no tears	Capture (6.1)	7 · 20 batches, inspect, re-test bypass

CED LECTURE 9: THREE KINDS OF SIMULATION

✓ **Analytical (slide 7)** ✓ **Virtual (slides 8–9)** ◑ **Physical: rig done (flow), tests 1–8 next**

Ready for TRIZ: contradictions and trade-offs

Each one comes from a number in this deck. The 39-parameter pairs are our first suggestion for the contradiction matrix; the principles are left for the session.

TECHNICAL CONTRADICTIONS (IMPROVE A, B GETS WORSE)

	Contradiction	Evidence	Suggested parameters	Principles
TC1	**Capture vs flow** A tighter sleeve catches 1–2 µm particles but cuts flow and sleeve life.	602 h: 89 % / 28 L/h / 71 L · G5: 78 % / 18 L/h / 27 L	#23 Loss of substance ↔ #39 Productivity	to do
TC2	**Speed vs capture certainty** Faster spin raises flow and sleeve life but drives the paper far above its rated velocity, and adds noise and unbalance.	1,500 rpm: 7.2× · 2,000: 12.9× · 2,500: 20×	#39 Productivity ↔ #27 Reliability (#31 harmful side effects)	to do
TC3	**Water speed vs simplicity** The water must turn with the basket, but vanes add a part to remove at every sleeve change and to clean.	no vanes 1.2 L/h · vanes 28 L/h	#9 Speed ↔ #33 Ease of operation (#36 Complexity)	to do
TC4	**Sample size vs sleeve life** Counting needs ≈ 20 L on one sleeve, but silt on the same sleeve shuts the flow first in dirty water.	borewell 18 L · turbid 4 L	#26 Quantity of substance ↔ #39 Productivity	to do
TC5	**Filter area vs size and mass** A bigger basket passes more water per batch but makes a bigger, heavier, costlier rotor.	Ø60 × 180 mm, 400 mL	#5 Area of moving object ↔ #1 Weight of moving object	to do
TC6	**All-steel wet path vs price** No plastic in the water path means deep-drawn and perforated stainless parts.	target ₹1,500–6,000	#31 Harmful side effects ↔ #32 Ease of manufacture	to do

PHYSICAL CONTRADICTIONS

PC1 **The sleeve's pores must be small (catch 1 µm) and large (pass 15 L/h).**
Separation idea: Space: coarse-to-fine layers; time: pre-filter, then sample sleeve

PC2 **The basket must spin fast (head, flow) and slowly (paper rating, noise).**
Separation idea: Time: fast start, slow finish; condition: speed set by drain time

PC3 **Particles must reach the paper (capture) and float inward (PP, PE > 20 µm).**
Separation idea: Condition: by density; place sampling where light ones land

PC4 **The sleeve must be fixed (seal, no bypass) and removable (sample, swap).**
Separation idea: Time: clamp only while spinning; system: sleeve in its own cartridge

TRADE-OFFS TO FIX WITH NUMBERS

- **Speed:** 1,000–1,750 rpm; choose with bench test 5.
- **Paper:** < 2 µm class; or two layers.
- **Batch:** 400 mL vs basket size and fill time.
- **Water:** tap vs borewell sleeve life.

Parameters from Altshuller's 39 (suggested; check in the session). Separation principles: space, time, condition, system

Ready for FMEA: 18 failure-mode candidates

Pahl & Beitz §10.4. Failure, effect and cause from this deck's numbers; occurrence, severity and detection are left blank for the team to rate (RN = O × S × D, > 125 critical).

#	Part / function	Failure mode	Effect on user	Cause	Evidence	O	S	D	Req.
1	**Sleeve seam and ends**	Water bypasses the paper	Unfiltered particles reach the jug	Seam or ends unsealed at ring and floor	Rig permeance 3.2× the datasheet				6.1
2	**Wet sleeve**	Tears or creases at 55 g	Bypass; fibres in the water	Wet strength over Ø2 holes, mesh not tight	Not tested				6.1
3	**Vanes**	Water lags (< 55 % of basket)	Flow under 15 L/h	Too few or short vanes, short spin	Rig: 6.5 % without vanes				2.1
4	**Sleeve paper**	Capture falls at high face velocity	1–2 µm particles pass	Rated at 1/7 of the working velocity	Model assumption				6.1
5	**Sleeve paper**	Blinds with silt	Flow < 15 L/h early	Borewell / turbid water	18 L / 4 L simulated				9.1
6	**Detection**	Cellulose fibres or stain read as plastic	False positives	Nile red uptake by cellulose	Bench note				6.2
7	**Detection**	Too few particles per field	Count meaningless	Sparse sample on a large sleeve	≈ 0.005 per field per batch				6.2
8	**Light polymers**	PP / PE miss the paper or land unevenly	Sample under-counts them	Inward drift > through-flow above ≈ 20 µm	Stokes drift, slide 13				1.2
9	**Basket**	Unbalance from an uneven wet sleeve	Vibration, noise, walking	Cake or water uneven, sleeve off-centre	5 g → 3.7 N				8.1
10	**Lid interlock**	Lid opened while spinning	Contact with the basket; spray	Switch fails or is bypassed	Design only				safety
11	**Lip seal**	Leaks into the dry base	Water on the motor and board	Wear at 1,500 rpm, dry running	Design only				3.5
12	**Bowl**	Spray over the rim	Wet bench; lost water	Water leaves the cage at ≈ 4 m/s	Skirt sized, untested				—
13	**Motor / driver**	Stalls or overheats with a full basket	No flow	Start-up torque, no current limit	Design only				2.1
14	**Speed control**	Speed drifts out of the band	Low flow or uncertain capture	No speed feedback (rig had none)	Team note, slide 5				2.1
15	**Sleeve change**	Sample touched or dropped	Contaminated sample; spilled catch	Four-step removal, wet hands	Not timed				10.2
16	**Spent sleeve**	Binned or rinsed	Microplastics back to the environment	No disposal route	—				10.1
17	**Cellulose sleeve**	Sheds fibres into the water	Particles in the output	Paper abrasion against the mesh	Not tested				3.2
18	**Cost**	Steel parts push the price over ₹6,000	Not affordable	Deep-drawn 304, perforated 316L, BLDC	Not costed				11.1

Next steps: measure the four assumptions, then TRIZ and FMEA

The rig is already built. Most of these tests need only vanes glued into it, a strobe app, beads and the Nile red kit we have for detection.

WEEK 1

Make the water spin

- Strobe or tracer: water speed now
- Glue 4 vanes into the rig; re-run A/B/C

WEEK 1

Find the bypass

- Dye through seam and ends
- Paper alone: K vs datasheet

WEEK 2

Finer paper on the rig

- Grade 602 h class sleeve
- Flow at 0 / 1,000 / 1,500 rpm

WEEK 2–3

Capture at speed

- Spiked 1, 2, 5, 10 µm beads
- Count on the sleeve and in the water

WEEK 3

Dirty water

- Borewell water: flow vs litres
- Fit α; blank-sleeve background

WEEK 4

TRIZ + FMEA session

- Rate the 18 candidates
- Matrix for TC1–TC6

BENCH TESTS: WHAT EACH ONE MEASURES, AND THE REQUIREMENT IT SETTLES

1. **Strobe / tracer on the water, with and without vanes**
   Measures the 85 % slip
   settles 2.1
2. **Dye through the sleeve seam and ends**
   Measures bypass, sealing
   settles 6.1
3. **Flow vs speed with a 602 h class sleeve**
   Measures the speed band
   settles 2.1
4. **Borewell run: flow vs litres**
   Measures cake α, sleeve life
   settles 9.1
5. **Spiked beads at 0 and 1,500 rpm**
   Measures capture at 7× face velocity
   settles 6.1
6. **Blank sleeve under Nile red**
   Measures background, fibres
   settles 6.2
7. **20 batches on one sleeve**
   Measures tears, unbalance, noise
   settles 3.5, 8.1
8. **Time a sleeve change**
   Measures swap ≤ 30 s
   settles 7.1